Team 16

Ambulance Booking System User Manual

12/10/2026

Document prepared by: Adithya Shyam D (241IT005), Dev Patel (241IT023), Ritesh Bhagat (241IT063)

Department of Information Technology, NITK Surathkal

Introduction

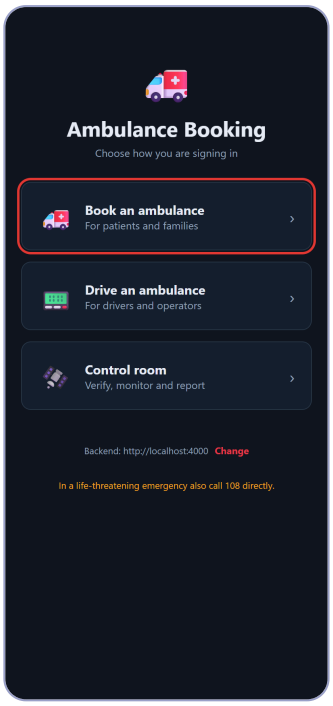
This manual describes the features of the Ambulance Booking System and explains, step by step, how each type of user works with it. The system is a single application with three modes:

- **Patient** (Book an ambulance): request an ambulance, track it live on a map, pay the invoice and rate the trip.
- **Driver / Operator** (Drive an ambulance): receive dispatch requests, accept them, and run the trip up to completion.
- **Control room** (Administrator): verify ambulances, monitor bookings and the fleet, and view reports and audit logs.

The same app runs on an Android phone (APK or Expo Go) and in a web browser, so the screens in this manual look the same on both.

Accessing the system

1. Open the **Ambulance Booking** app on your phone, or open the web version in a browser.
2. On the first screen, choose how you are signing in: **Book an ambulance**, **Drive an ambulance** or **Control room**.



Mode selection screen

> In a life-threatening emergency, also call **108** directly. The app shows this reminder on the sign-in and booking screens.

Actions available to each user

Patient

- Register, sign in with a password, or sign in with a one-time code (OTP).
- Book an ambulance:
  - Set the pickup location (GPS, map pin or address search)
  - Choose the ambulance type:
    - BLS – Basic Life Support
    - ALS – Advanced Life Support
    - ICU – Mobile ICU
    - PTV – Patient Transport
  - Add patient details and see an estimated fare
- Track the assigned ambulance live, with its ETA and trip progress.
- Call the driver, or cancel the request before the patient is on board.
- View past trips, pay the invoice (UPI, Card, Netbanking or Cash) and rate the trip.
- Edit their profile and read notifications.

Driver / Operator

- Register as an individual driver or an organisation, and submit ambulance details for verification.

- Go on duty or off duty.
- Accept or decline dispatch requests.
- Step through the trip: on the way, arrived, patient on board, trip completed.
- View earnings, trip history, ratings and their ambulance profile.

Control room (Administrator)

- Monitor live dispatch figures on the overview dashboard.
- Search and filter bookings by status; view booking details; re-dispatch or cancel a booking.
- View every ambulance on the fleet map.
- Verify, reject or suspend ambulance providers.
- View reports, the WhatsApp/SMS alert log, the audit trail, and manage user accounts.

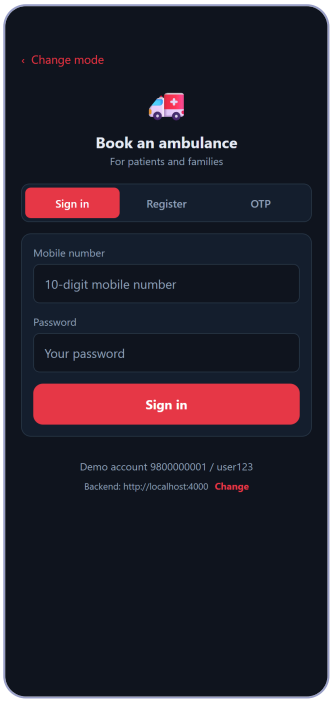
Contents

1. Signing in and registration (all users)
2. Patient features
3. Driver / Operator features
4. Control room features
5. Appendix: demo accounts

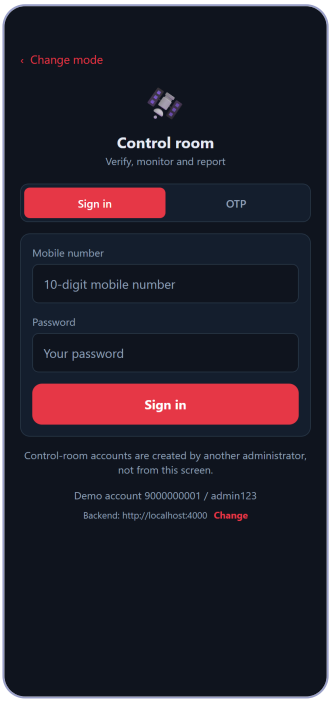
1. Signing in and registration

1.1 Sign in with a password

- After choosing a mode, the **Sign in** tab is shown.
- Enter your 10-digit **Mobile number** and **Password**, then click **Sign in**.
- To go back and choose a different mode, click ‹ **Change mode** at the top.



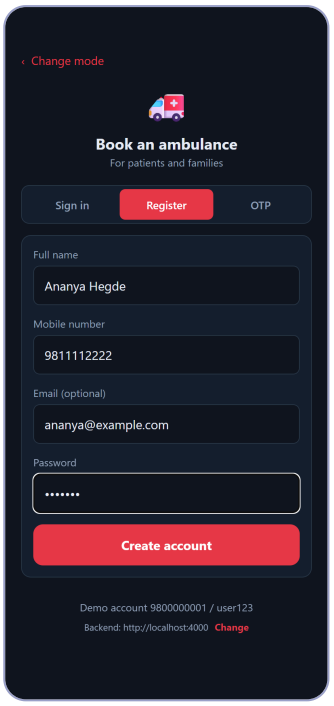
Sign in (patient)



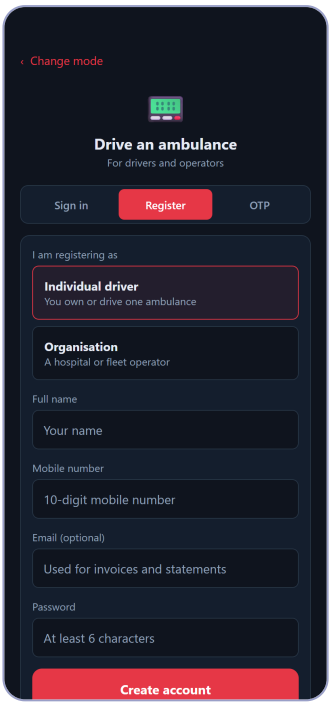
Sign in (control room)

1.2 Register a new account

- Click the **Register** tab and enter:
  - Full name
  - Mobile number
  - Email (optional, used for invoices)
  - Password (at least 6 characters)
- Drivers must also choose **Individual driver** or **Organisation** (a hospital or fleet operator).
- Click **Create account**.



Patient registration

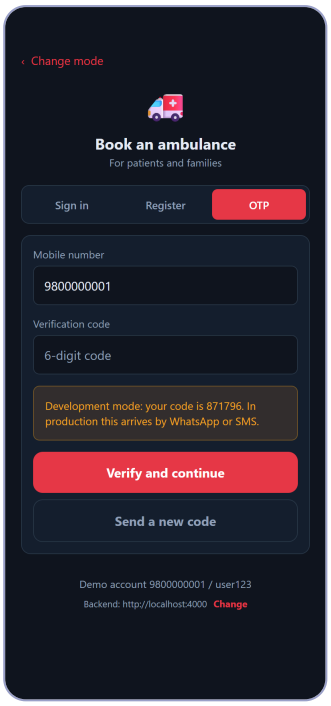


Driver / organisation registration

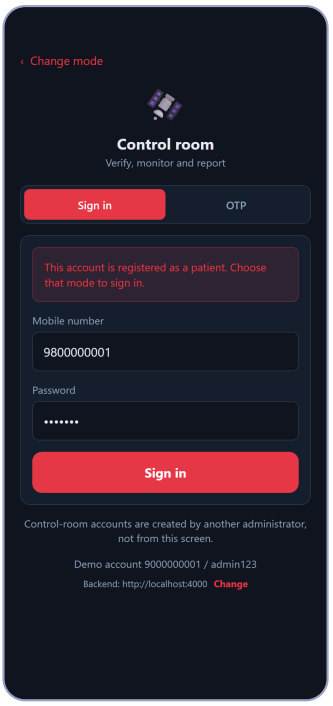
> Control-room accounts cannot be created from the sign-in screen. Only an existing administrator can create one.

1.3 Sign in with a one-time code (OTP)

- Click the **OTP** tab, enter your mobile number and click **Send code**.
- Enter the 6-digit **Verification code** you receive by WhatsApp or SMS, then click **Verify and continue**.
- Click **Send a new code** if the code did not arrive.



OTP sign in (in development mode, the code is shown on screen)



Signing in under the wrong mode is refused

> Each account belongs to one role. If you sign in under the wrong mode, for example a patient account under **Control room**, the app tells you which mode to choose instead.

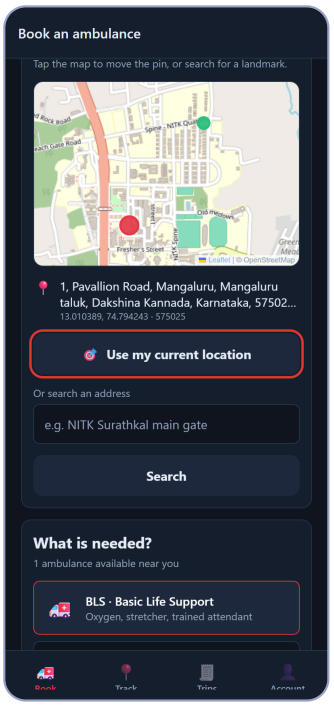
> **Backend address:** the line *Backend: http://… Change* on the sign-in screen shows which server the app is connected to. If the app cannot connect, click **Change**, enter the server address given by your administrator, and save.

2. Patient features

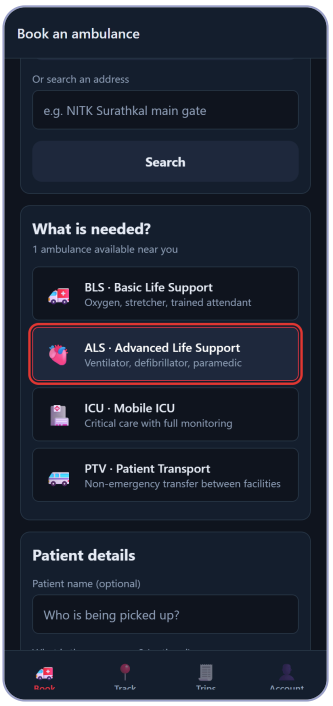
After signing in as a patient, four tabs appear at the bottom of the screen: **Book**, **Track**, **Trips** and **Account**.

2.1 Booking an ambulance

- **Set the pickup location.** Under *Where are you?*, the app reads your GPS position and places a red pin on the map. You can also:
  - click **Use my current location** to read the GPS again,
  - tap anywhere on the map to move the pin, or
  - type a landmark in *Or search an address*, click **Search**, and pick a result.
- Green dots on the map are ambulances that are free near you.



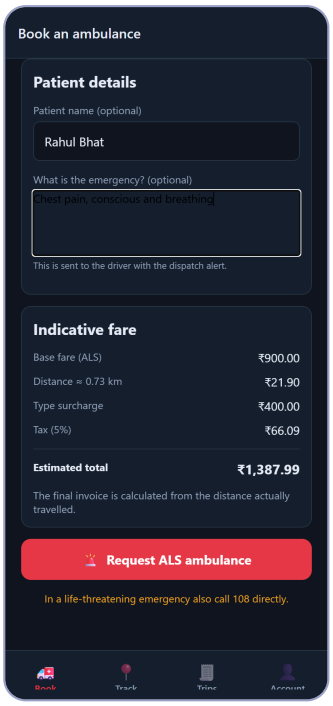
Setting the pickup location



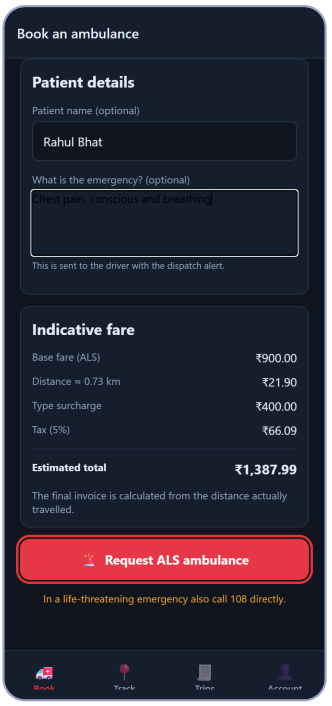
Choosing the ambulance type

- **Choose the ambulance type** under *What is needed?*:
  - **BLS**: Basic Life Support (oxygen, stretcher, trained attendant)
  - **ALS**: Advanced Life Support (ventilator, defibrillator, paramedic)
  - **ICU**: Mobile ICU (critical care with full monitoring)
  - **PTV**: Patient Transport (non-emergency transfer between facilities)
- **Enter the patient details** (both optional):
  - Patient name
  - What is the emergency? The driver receives this text with the dispatch alert.
- Check the **Indicative fare**: base fare, distance charge, type surcharge, tax and estimated total. The final invoice uses the distance actually travelled.

- Click **Request <type> ambulance** and confirm the pickup address in the dialog that opens.



Patient details

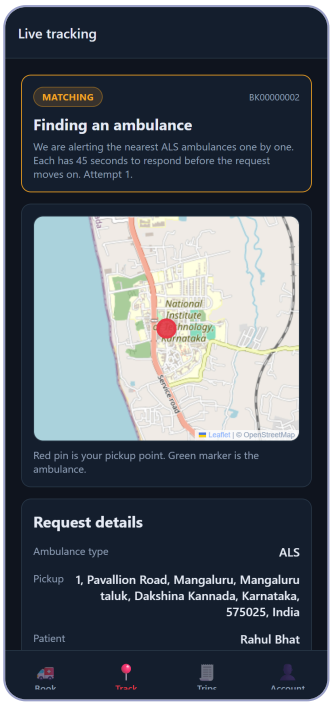


Indicative fare and the request button

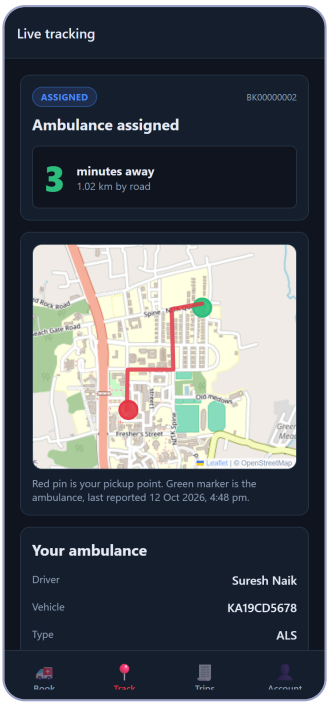
- The request is created and the app opens the **Track** tab.

2.2 Tracking the ambulance

- While the status is **MATCHING**, the system offers your request to the nearest suitable ambulance. Each driver has 45 seconds to respond. If a driver declines or does not answer, the next ambulance is offered automatically.
- Once a driver accepts, the status changes to **ASSIGNED**. The screen then shows:
  - how many minutes away the ambulance is, and the road distance,
  - a live map (red pin: pickup point, green marker: the ambulance) and the route.

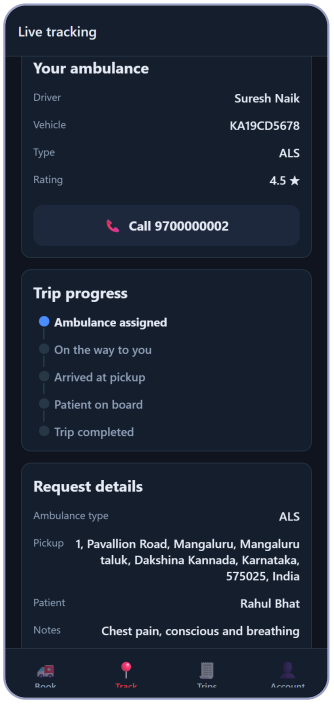


Finding an ambulance

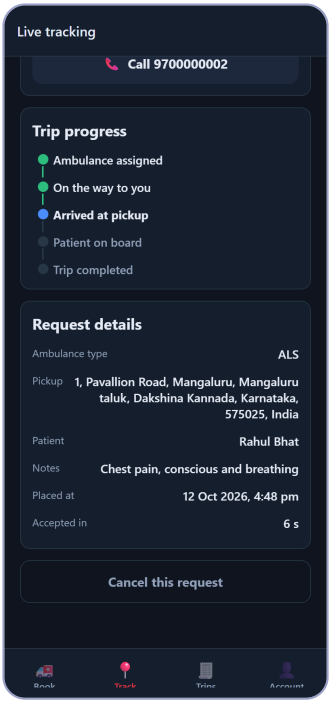


Ambulance assigned, with live ETA

- Scroll down to see **Your ambulance** (driver name, vehicle number, type, rating). Click **Call <number>** to phone the driver.
- **Trip progress** updates as soon as the driver taps each step: Ambulance assigned → On the way to you → Arrived at pickup → Patient on board → Trip completed.



Ambulance and trip details

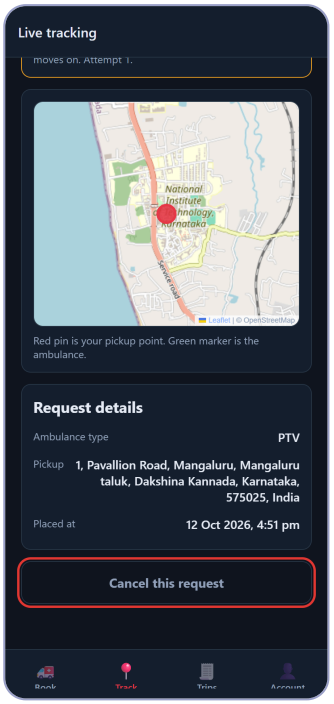


Trip progress updating live

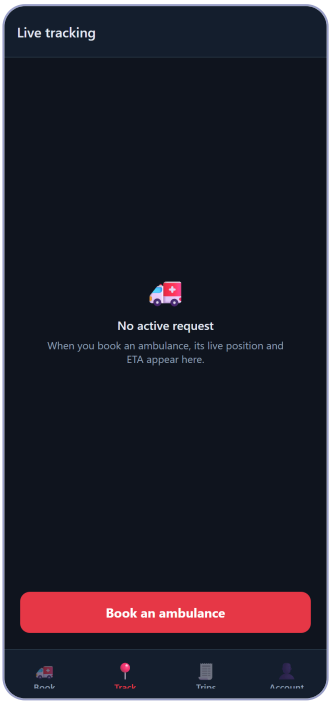
2.3 Cancelling a request

- On the **Track** tab, scroll down and click **Cancel this request**, then confirm.

- You can cancel at any time until the patient is on board.
- The request is cancelled, the driver (if any) is told, and the Track tab returns to *No active request*.



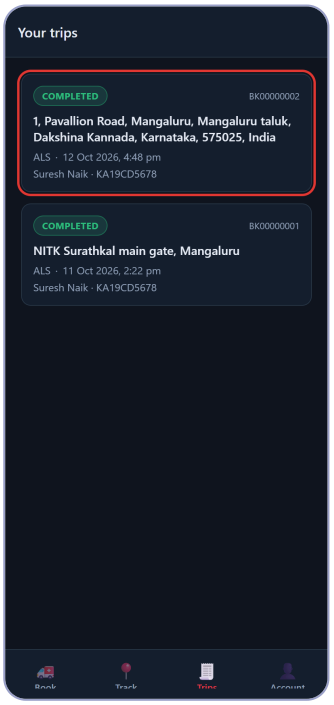
Cancel this request



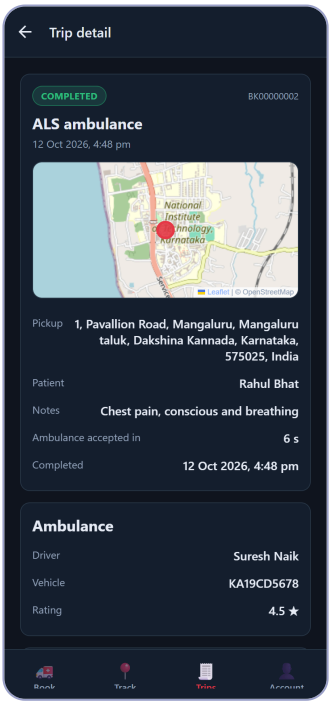
Request withdrawn

2.4 Viewing past trips

- Open the **Trips** tab to see all your bookings, newest first, with their status, booking ID, pickup address, driver and vehicle.
- Click a trip to open **Trip detail**: status, pickup, patient, notes, response time, ambulance and invoice.



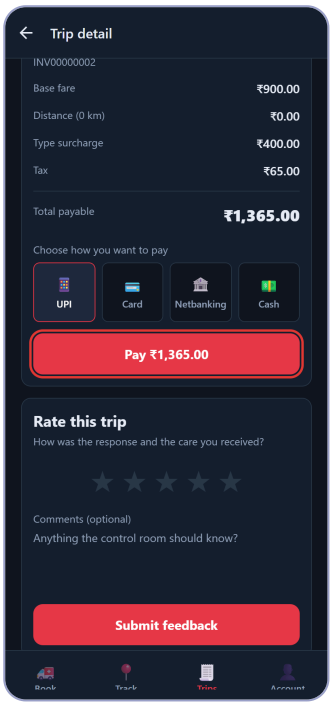
Your trips



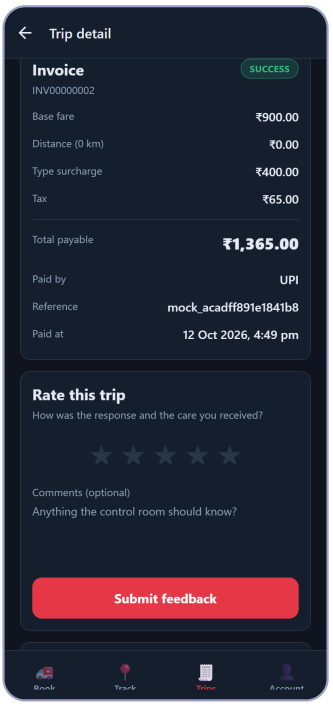
Trip detail

2.5 Paying the invoice

- When a trip is completed, an invoice is raised. Open the trip from **Trips**, or click **View invoice and rate the trip** on the Track tab.
- Under **Invoice**, check the base fare, distance charge, type surcharge, tax and **Total payable**.
- Choose how you want to pay: **UPI**, **Card**, **Netbanking** or **Cash**, then click **Pay ₹....**
- The invoice changes to **SUCCESS** and shows the payment mode, transaction reference and time.



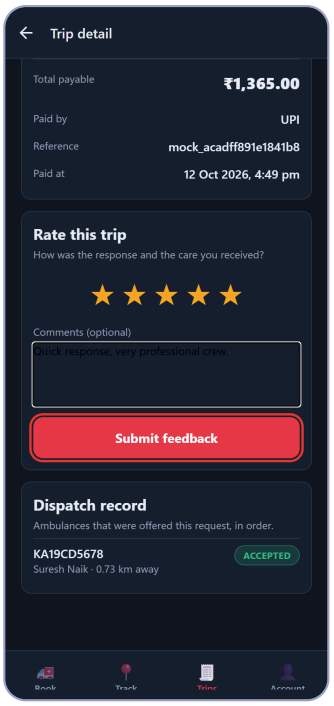
Choosing a payment mode



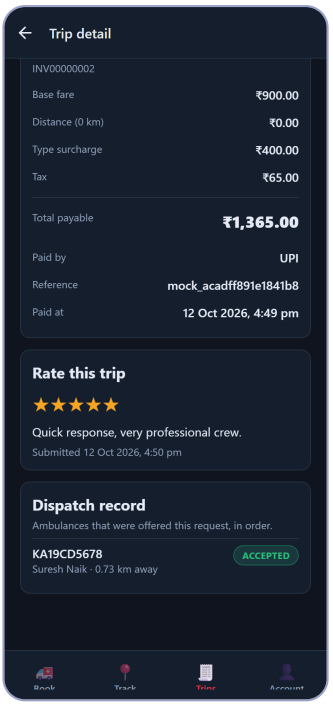
Invoice paid

2.6 Rating the trip

- Under **Rate this trip**, tap 1 to 5 stars and optionally add a comment.
- Click **Submit feedback**. The rating is added to the driver's profile and shown to the control room.
- **Dispatch record** lists every ambulance that was offered the request, in order.



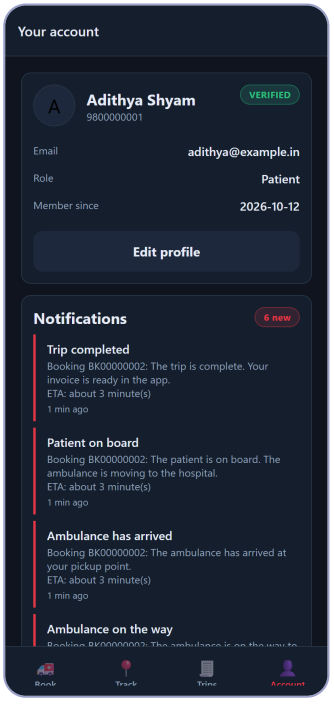
Rating the trip



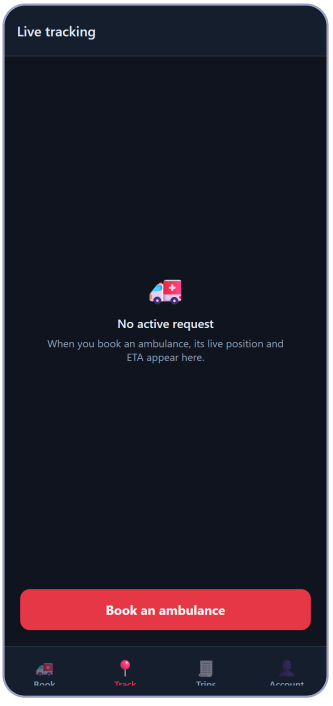
Feedback submitted

2.7 Account and notifications

- The **Account** tab shows your name, mobile number, email, role and member-since date.
- Click **Edit profile** to change your name or email, then click **Save**.
- **Notifications** lists every status update for your bookings. Click **Mark all as read** to clear the count.
- Click **Sign out** at the bottom to log out.



Your account



Track tab with no active request

3. Driver / Operator features

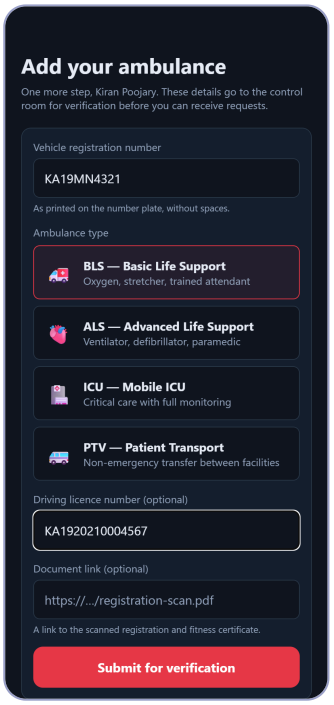
After signing in under **Drive an ambulance**, four tabs appear: **Console**, **Trip**, **Earnings** and **Profile**.

3.1 Adding your ambulance

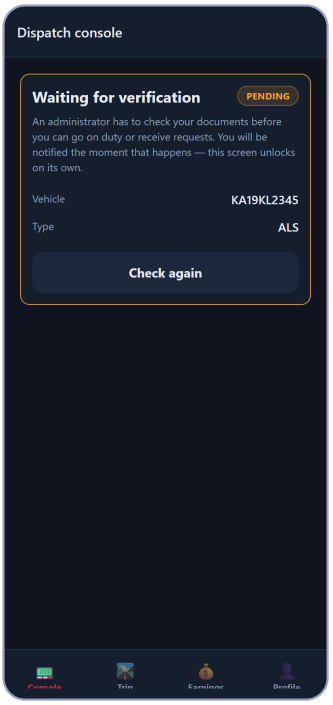
- A newly registered driver is first asked to **Add your ambulance**:
  - Vehicle registration number (as on the number plate, without spaces)
  - Ambulance type: BLS, ALS, ICU or PTV
  - Driving licence number (optional)
  - Document link (optional): a link to the scanned registration and fitness certificate
  - Organisation name and address (organisation accounts only)
- Click **Submit for verification**.

3.2 Waiting for verification

- Until the control room verifies your documents, the console shows **Waiting for verification** with status **PENDING**. You cannot go on duty or receive requests.
- The console unlocks automatically as soon as you are verified. You can also click **Check again**.



Add your ambulance

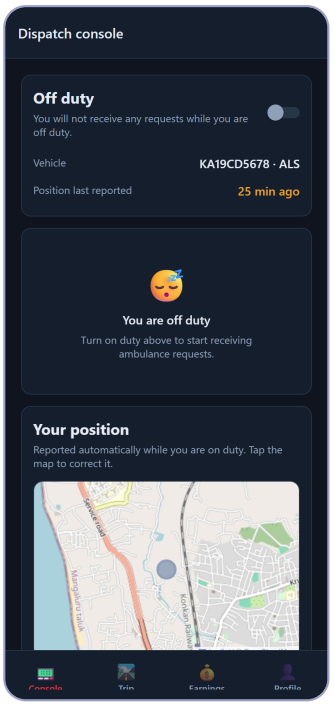


Waiting for verification

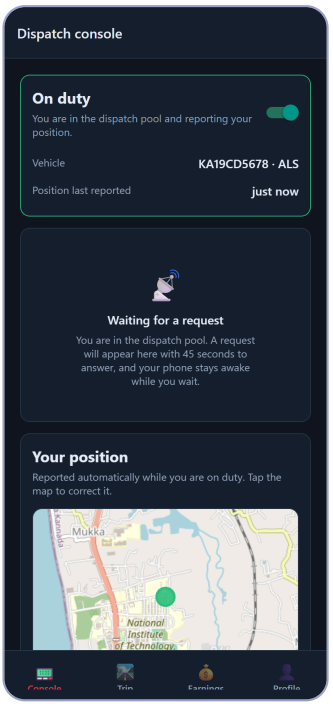
3.3 Going on duty

- On the **Console** tab, turn on the switch next to **Off duty**. The card turns green and shows **On duty**.
- While on duty, the app reports your position every 20 seconds and you are in the dispatch pool. Keep the app open.

- Turn the switch off to stop receiving requests.



Off duty

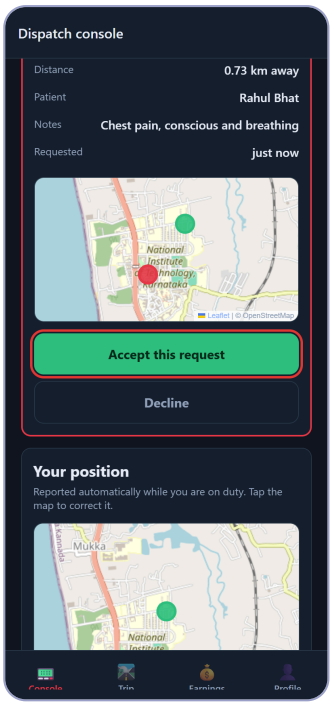


On duty, waiting for a request

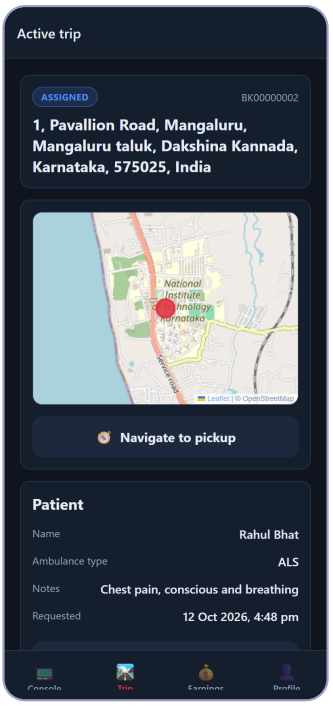
> If the app stops reporting its position for 15 minutes, the system takes you off duty automatically and tells you why.

3.4 Accepting or declining a request

- A new request appears on the console with a 45-second countdown. The same alert is also sent to you by WhatsApp.
- The request shows the ambulance type, pickup address, distance, patient name, notes and a map.
- Click **Accept this request** to take the job, or **Decline** to pass it on. If you do not respond within 45 seconds, the request goes to the next ambulance.



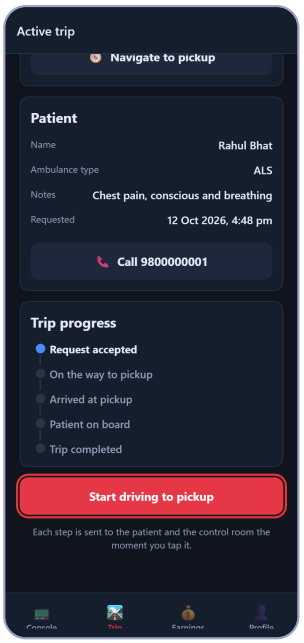
Incoming request



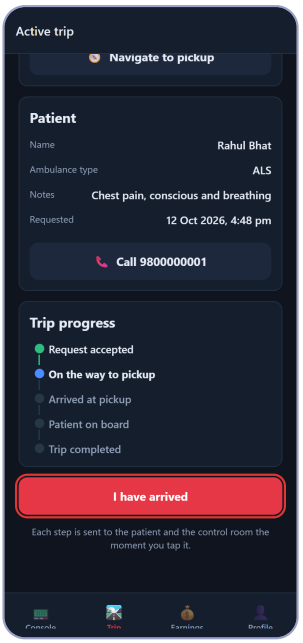
Active trip after accepting

3.5 Running the trip

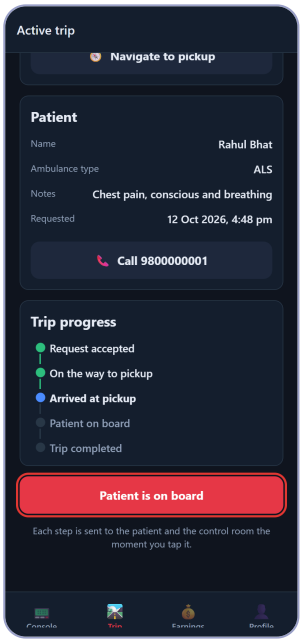
- Open the **Trip** tab. It shows the pickup on a map, **Navigate to pickup** (opens turn-by-turn directions), the patient details and **Call <number>**.
- The screen always offers exactly one button: the next step. Tap it when that step happens:
  1. **Start driving to pickup**
  2. **I have arrived**
  3. **Patient is on board**
  4. **Complete the trip** (asks for confirmation)
- Each step is sent to the patient and the control room immediately.



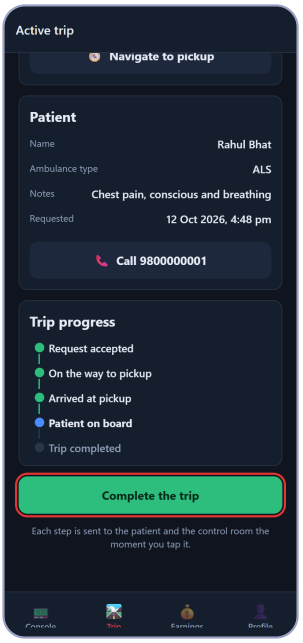
Step 1



Step 2



Step 3



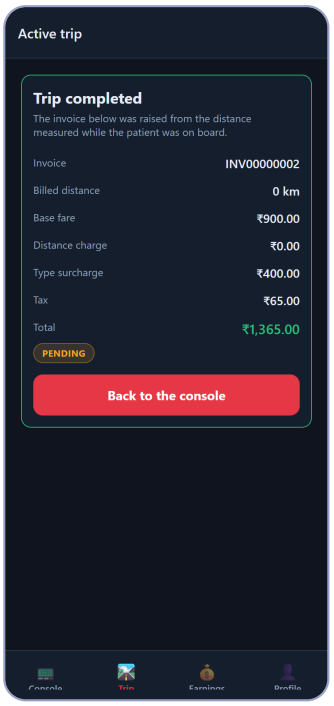
Step 4

3.6 Trip completion and invoice

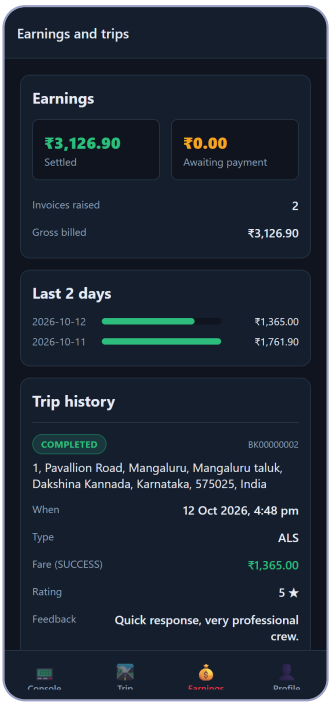
- When the trip is completed, the invoice is generated automatically. The fare is calculated from the distance the server measured while the patient was on board.
- The invoice shows billed distance, base fare, distance charge, type surcharge, tax and total, with payment status **PENDING** until the patient pays.
- Click **Back to the console** to wait for the next request.

3.7 Earnings and trip history

- The **Earnings** tab shows settled and awaiting-payment amounts, invoices raised, gross billed, a bar for each recent day, and every trip with its fare, rating and feedback.



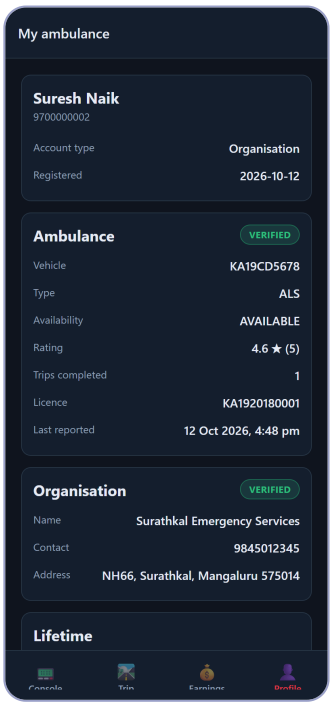
Trip completed and invoice



Earnings and trips

3.8 Profile

- The **Profile** tab (*My ambulance*) shows your account, ambulance (vehicle, type, availability, rating, trips completed, licence), organisation and lifetime statistics.
- Click **Sign out** at the bottom to log out.



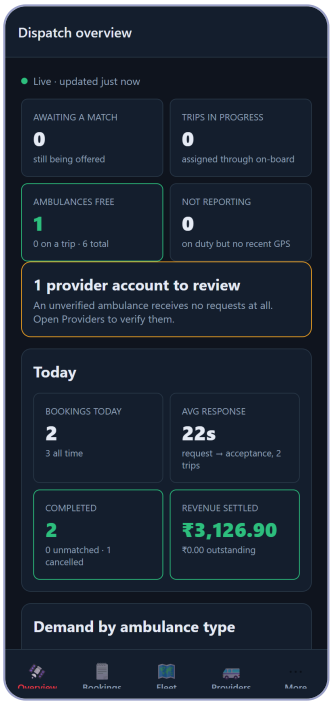
My ambulance

4. Control room features

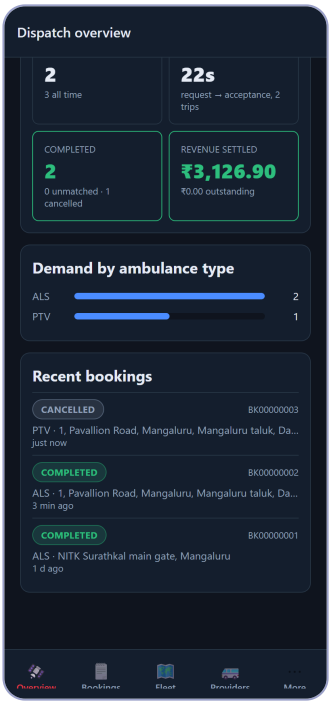
After signing in under **Control room**, five tabs appear: **Overview**, **Bookings**, **Fleet**, **Providers** and **More**. The control room is usually opened in a browser on a laptop.

4.1 Dispatch overview

- Live counts that refresh automatically:
  - Awaiting a match: requests still being offered
  - Trips in progress: from assigned through on board
  - Ambulances free
  - Not reporting: on duty but no recent GPS position
- A banner shows how many provider accounts are waiting for review.
- **Today**: bookings today, average response time, completed trips and revenue settled.
- **Demand by ambulance type** and **Recent bookings**.



Dispatch overview

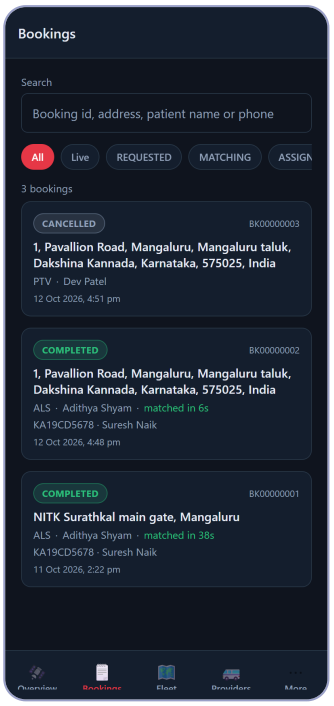


Today and demand by type

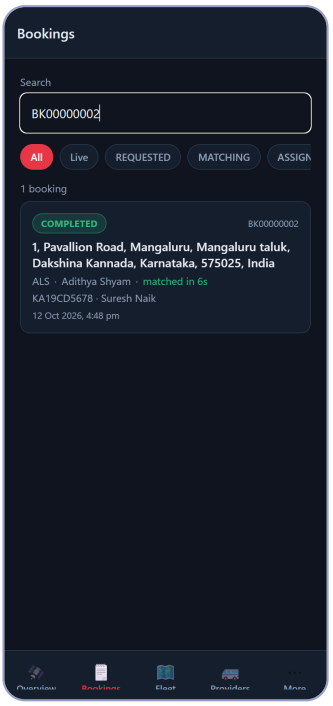
4.2 Searching for bookings

- Open the **Bookings** tab. Type a keyword in the **Search** box.
  (Note: you can search by booking ID, address, patient name or phone number.)
- Filter by status using the chips below the search box:
  - All
  - Live
  - REQUESTED, MATCHING, ASSIGNED, ENROUTE, ARRIVED, ONBOARD
  - COMPLETED, CANCELLED, FAILED

- For example, after entering *BK00000002* in the search box:



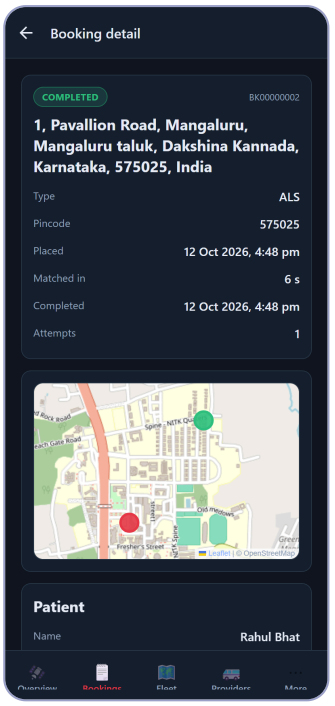
All bookings



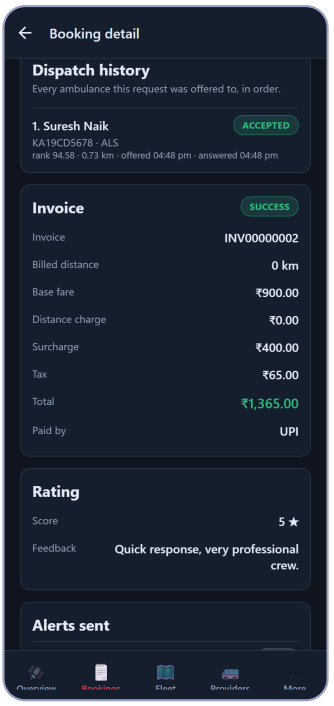
Search result

4.3 Viewing booking details

- Click any booking to open **Booking detail**: status, pickup, type, time placed, match time, number of attempts, map and patient (with **Call the patient**).
- Scroll down for **Dispatch history** (every ambulance offered, with the time offered and answered), **Invoice**, **Rating** and **Alerts sent**.
- For a request that has failed or is still matching, click **Re-dispatch this request** to offer it again.
- For an active booking, click **Cancel this booking** to cancel it.



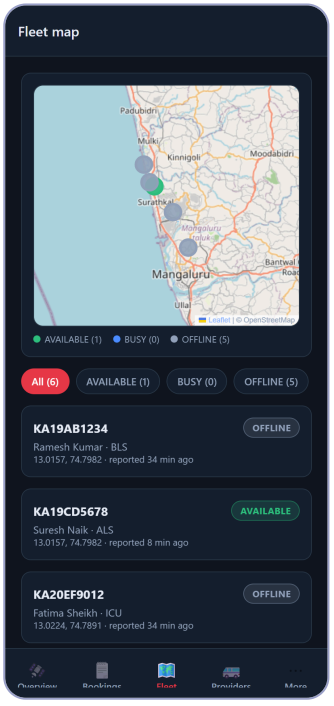
Booking detail



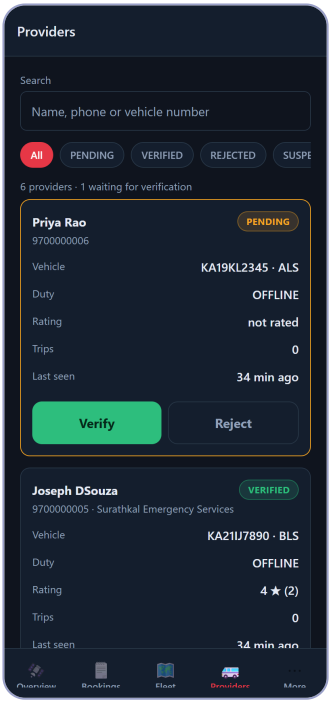
Dispatch history, invoice and rating

4.4 Fleet map

- The **Fleet** tab shows every ambulance's last reported position on a map, coloured by availability (available, busy, offline).
- Filter the list below the map by **All**, **Available**, **Busy** or **Offline**.



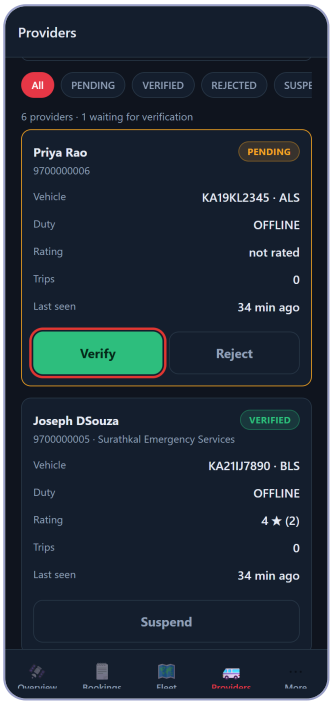
Fleet map



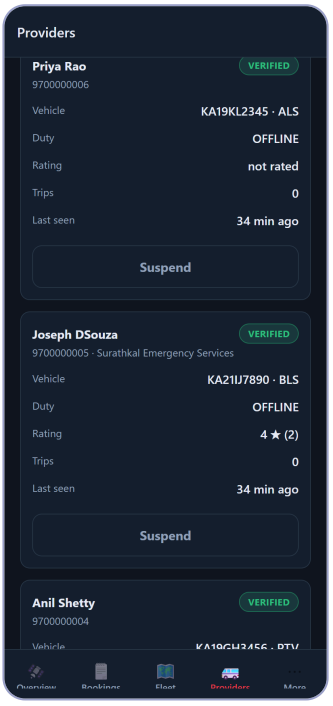
Providers list

4.5 Verifying ambulance providers

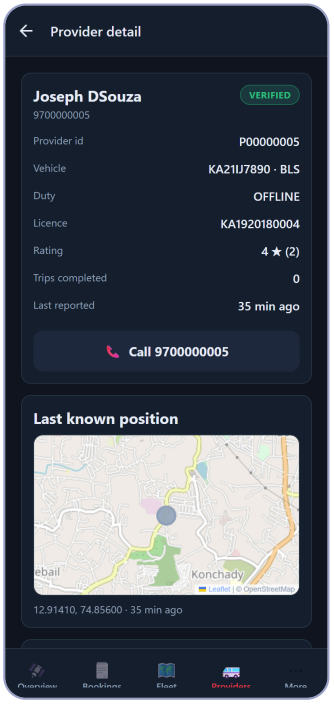
- Open the **Providers** tab. Search by name, phone or vehicle number, or filter by PENDING, VERIFIED, REJECTED or SUSPENDED.
- For a provider with status **PENDING**, check the details and click **Verify** or **Reject**.
- The provider's status changes to **VERIFIED**, and their driver console unlocks immediately.
- A verified provider can be suspended with **Suspend**.
- Click a provider to open **Provider detail**: provider ID, vehicle, duty status, licence, rating, trips, **Call** button and last known position.



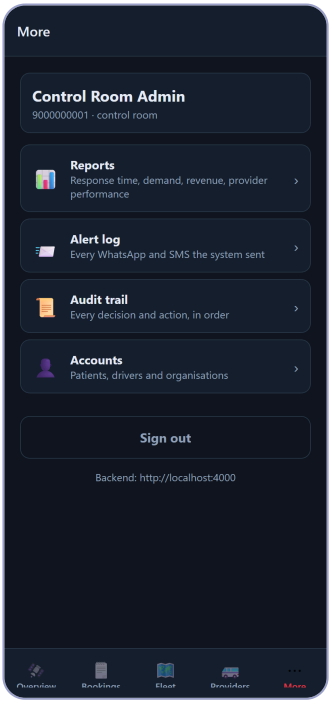
Verifying a pending provider



Provider verified



Provider detail



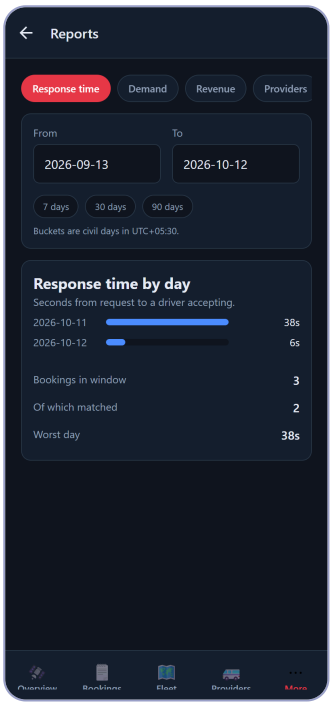
More menu

4.6 Reports

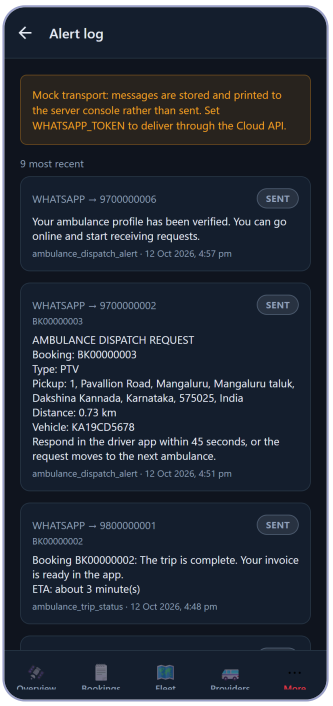
- Open **More** → **Reports**.
- Choose a report: **Response time**, **Demand**, **Revenue** or **Providers**.
- Set the **From** and **To** dates, or pick 7, 30 or 90 days. Days are counted in Indian Standard Time.

4.7 Alert log

- **More** → **Alert log** lists every WhatsApp and SMS message the system sent (dispatch alerts, OTPs, trip status updates), with its delivery state.



Reports



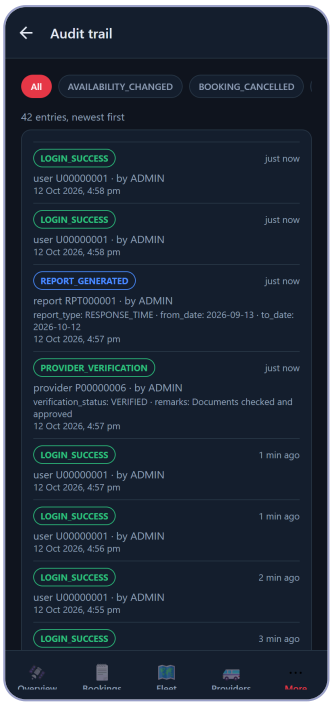
Alert log

4.8 Audit trail

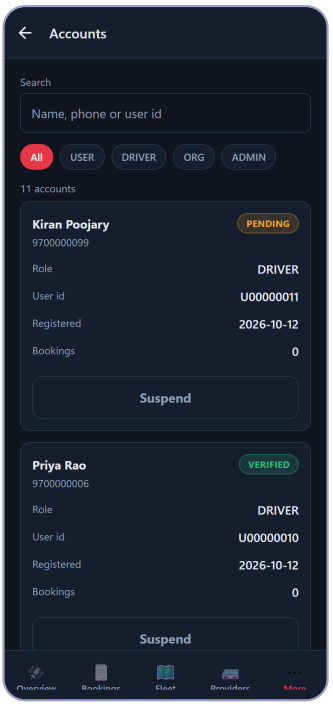
- **More** → **Audit trail** records every decision and action in order: sign-ins, bookings, dispatch offers, status changes, cancellations, provider verification and reports. Filter by action type using the chips at the top.

4.9 Managing accounts

- **More** → **Accounts** lists all patients, drivers, organisations and administrators. Search by name, phone or user ID, or filter by role.
- Click **Suspend** to block an account from signing in, or **Restore** to allow it again.



Audit trail



Accounts

5. Appendix: demo accounts

The demo database includes these accounts for testing. Each one must be used with the mode shown.

Role	Mobile	Password	Mode to choose
Administrator	9000000001	admin123	Control room
Patient	9800000001	user123	Book an ambulance
Patient	9800000002	user123	Book an ambulance
Driver (ALS)	9700000002	driver123	Drive an ambulance
Driver (ICU)	9700000003	driver123	Drive an ambulance
Driver (PTV)	9700000004	driver123	Drive an ambulance
Driver (unverified)	9700000006	driver123	Drive an ambulance